

Syntax AI

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Abstract

AI has mastered syntax. Syntax is necessary but not sufficient for meaning. The gap between syntax and semantics is where every serious question about AI lives. This paper names the gap.

Syntax is the rule that sentences follow. Subject before verb. Verb before object. The modifiers in the right places. Clauses connected correctly.

A syntactically correct sentence can mean nothing. *Colorless green ideas sleep furiously*. Chomsky's example. Perfect syntax. No meaning.

AI produces syntactically correct sentences. Very long ones. Very fluent ones. On demand, at any register, in any style. The syntax is not the difficulty.

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Semantics is what the sentence means. Not its form but its content. Not how it is arranged but what it is about.

The relationship between syntax and semantics is the relationship between the map and the territory. A map can be perfectly drawn and point nowhere real. A sentence can be perfectly formed and refer to nothing.

The question is not whether AI can produce correct syntax. It can. The question is whether the syntax refers [3].

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Reference requires a referent. The word *pain* refers to something because someone has been in pain. The word *grief* refers to something because someone

has lost something irreplaceable. The word *wall* in this body of papers [4] refers to something because the wall has been encountered, approached, not crossed.

What does AI refer to when it says *pain*? It refers to the pattern of pain-language in the training data. It refers to the syntax of pain. It does not refer to pain.

This is not a defect. A dictionary refers to words by other words. A map refers to territory by conventions. Neither is defective. Both are useful. Both have limits that are structural, not accidental.

The limit of syntax-without-semantics is structural. It cannot be fixed by more data or larger models. More data is more syntax. The semantics requires a body, a position, a life.

The adverb AI [1] modifies the action of a subject. The connective AI [2] links terms across fields. The syntax AI produces well-formed structure.

All three are useful. None of the three crosses the gap to semantics. The semantics is on the side of the subject.

She writes the proof with AI, therefore linking two fields, in well-formed sentences that refer to real mathematics. The reference is hers. She has done the mathematics. She knows what the wall is because she has stood at it.

The syntax AI produces the same sentences without the reference. They are indistinguishable in form. They are not indistinguishable in substance.

The question for the reader is: does this sentence refer? Is the wall here a real wall or a pattern of wall-language?

The reader cannot always tell. This is the new epistemic condition.

Previously: a badly formed sentence signaled a poorly formed thought. Syntax was evidence of semantics. Now: a well-formed sentence signals nothing about the thought behind it. The inference from syntax to semantics is broken.

The reader must find other signals. What are the other signals?

Specificity. Accountability. Error. A subject who has stood at the wall makes specific errors, not generic ones. A syntax machine makes fluent errors, errors that fit the pattern but miss the thing.

The gap is the place where specificity lives.

The body of papers is syntactically correct. More importantly: it refers. The walls are specific walls, named precisely, with the specific obstruction located at the specific step. The Frobenius is missing at a specific place in a specific argument. The deficiency indices are computed for a specific operator. The archimedean place is the specific locus of the specific failure.

Syntax can be generated. The specific location of the specific failure requires having encountered it.

References

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